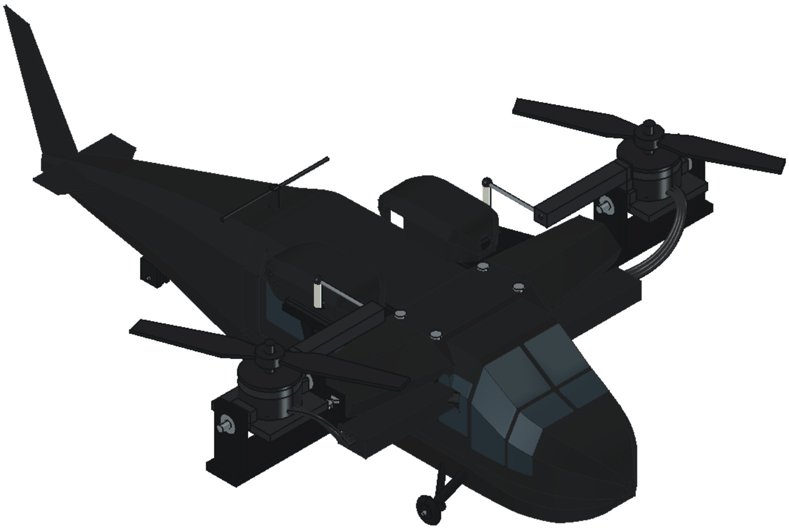
Build & engineering guide

KESTREL MINI R4

Black Hawk inspired styling. Two independently tilting lift pods.



NOMINAL MASS	WITH 10% GROWTH	INSTALLED PRINTS
665 g	731 g	40 pieces

Design / fit prototype - flight qualification remains open. The CAD includes electronics, wiring and a lightweight tail. Published motor data support takeoff in theory under the stated assumptions; the aircraft has not been built or flown.

Revision R4 | 9 September 2026 | Units: mm, g, N and N m. Baseline uses a 4S1300 LiPo, commercial 5-inch props and no optional guards.

Purchase list / electronics

No hardware is owned. These selections establish the CAD and calculation basis. Buy only after resolving the marked interfaces; a similar-looking substitute invalidates the existing fit or thrust evidence. Reference IDs link to manufacturer pages on page 13.

Qty	Selected part	Total g	Key fit / selection requirement
2	EMAX RS2205 2300KV motor [MOTOR]	60.0	Genuine original RS2205; 16 x 19 mm M3 mounting. Verify lead exit, shaft and screw engagement.
2	HQProp HQ5045BN two-blade propeller [MOTOR]	6.0	One CW and one CCW; exact blade from thrust table. Buy commercial props; never print these.
2	EMAX Lightning 30A ESC [ESC]	16.0	Legacy BLHeli; 27 x 12 x 5 mm model reserve. Confirm actual dimensions, cooling and supported input protocol.
2	Blue Bird BMS-115WV+ tilt servo [SERVO]	24.0	6 V; 23.2 x 10 x 23 mm body. Catalog 11 +/-0.5 g; 12 g planning allowance. Ears/spline stack provisional.
1	Matek F405-TE flight controller [FC]	10.0	36 x 46 mm; verify 30.5 mm mount spacing and exact board variant. Bicopter firmware support required.
1	Hobbywing UBEC 5A [BEC]	21.0	50 x 17 x 10 mm; set to 6 V for servos only. 5 A continuous rating; actual simultaneous servo demand unknown.
1	RadioMaster RP1 V2 ELRS 2.4 GHz receiver [RX]	2.2	13 x 11 x 3 mm, antenna included in allocation. Requires compatible ELRS transmitter; FlySky AFHDS radio is not compatible.
1	4S 1300 mAh LiPo - exact SKU OPEN [BATTERY]	170.0	76 x 38 x 31 mm maximum planning envelope, including stock leads in mass. Must support measured pack current and fit; no C-label guarantee.
1	XT60 aircraft-side connector	4.0	Mate to battery connector; insulated solder cups and strain relief. Stock battery half is counted with battery.
1	470 uF 35 V low-ESR capacitor	3.0	Reserve diameter 10 x 16 mm; confirm ripple rating and polarity. Lead length affects suppression.
1	5 V active buzzer	2.0	Reserve diameter 10 x 5 mm; verify compatibility with FC buzzer output and polarity.

Battery selection is OPEN. The 170 g allocation includes stock leads; confirm actual pack current capability, mass and dimensions. Four 21700 cells would put this layout near 799 g before contingency and require packaging changes.

Mechanical parts & bench equipment

Qty	Purchased mechanical part	Total g	Interface requirement
4	MR104 bearing, 4 x 10 x 4 mm	5.2	Metal bearings; confirm fit, shields, axial support and smooth rotation.
4	4 mm metal trunnion / shoulder pin	8.0	Nominal 14 mm modeled length. Final shoulders, spacers, retaining method and bearing stack OPEN.
2	OEM-compatible 25T servo horn	1.6	14 mm effective pivot-to-ball radius. Verify spline and screw; do not substitute a loose printed spline.
2	M2 pushrod with ball-link ends	4.0	35 mm center-to-center modeled link; four ball joints total. Verify travel, clearance and attachment geometry.

Installation allowances - 77 g total

Budget	Included material	g
Harness	Wire, solder, secondary connectors, insulated power splice	36
Hardware	Other screws, nuts, spacers, isolators, axles and retainers	25
Installation	Straps, foam, ties and insulation	8
Finish	Adhesive and restrained matte-black finish	8

Modeled fastener dimensions: eight motor M3 x7, four FC M3 x10, four beam/keel M3 x22, four yoke M3 x10; two diameter-2 x12 mm main axles and one diameter-2 x9 mm tail axle. Lengths require received-hardware verification. Motor screw ends must clear windings. Tail splice, shell closure, servo ear screws and trunnion retention are not fully detailed.

Ground equipment - outside aircraft mass

• ELRS 2.4 GHz transmitter/module compatible with RP1; the initial FlySky radio suggestion does not match this receiver.

• 4S-compatible LiPo balance charger and supply; scale, calipers, multimeter, current/voltage logging and soldering tools.

• Restrained thrust and actuator test fixtures, vibration measurement, fit coupons and appropriate filament/slicer.

Purchased component allocations total 337 g. Do not count supplied horns, link ends, battery connectors or stock leads twice. Exact hardware is still a fit and qualification task.

Print inventory / 1 of 2

One of each listed STL. Names correspond exactly to files in stl/. Separated glazing pieces and mirrored parts have individual files. CAD masses use full material volume at 1.24 g/cm^3; confirm slicer mass and actual print weight.

STL name (without .stl)	CAD g	Bounding box, mm
P01_Battery_keel	35.28	136.0 x 58.0 x 48.0
P02_Crossbeam	45.18	73.0 x 168.0 x 19.0
P03_Avionics_deck	6.30	54.0 x 49.0 x 4.0
P04_Tail_spine	3.15	163.3 x 12.0 x 35.5
P05_Horizontal_tailplane	1.64	31.0 x 48.0 x 1.2
P05_Vertical_tail_fin	2.19	52.0 x 1.3 x 70.0
P06_Cabin_upper	25.70	210.0 x 62.0 x 51.0
P07_Cabin_lower	22.95	208.8 x 62.0 x 17.0
P08_Windshield	1.45	20.0 x 29.9 x 40.0
P08_Windshield_segment_2	0.74	12.5 x 28.8 x 26.3
P08_Windshield_segment_3	1.45	20.0 x 29.9 x 40.0
P08_Windshield_segment_4	0.74	12.5 x 28.8 x 26.3
P09_Cabin_side_glazing_0	0.53	18.0 x 4.5 x 27.0
P09_Cabin_side_glazing_0_segment_2	1.10	28.0 x 4.8 x 28.0
P09_Cabin_side_glazing_0_segment_3	1.03	28.0 x 5.3 x 28.0
P09_Cabin_side_glazing_1	0.53	18.0 x 4.5 x 27.0
P09_Cabin_side_glazing_1_segment_2	1.10	28.0 x 4.8 x 28.0
P09_Cabin_side_glazing_1_segment_3	1.03	28.0 x 5.3 x 28.0
P10_Main_gear_1	1.28	27.7 x 17.5 x 26.7
P10_Main_gear__1	1.28	27.7 x 17.5 x 26.7

Print inventory / 2 of 2

One of each listed STL. Names correspond exactly to files in stl/. Separated glazing pieces and mirrored parts have individual files. CAD masses use full material volume at 1.24 g/cm^3; confirm slicer mass and actual print weight.

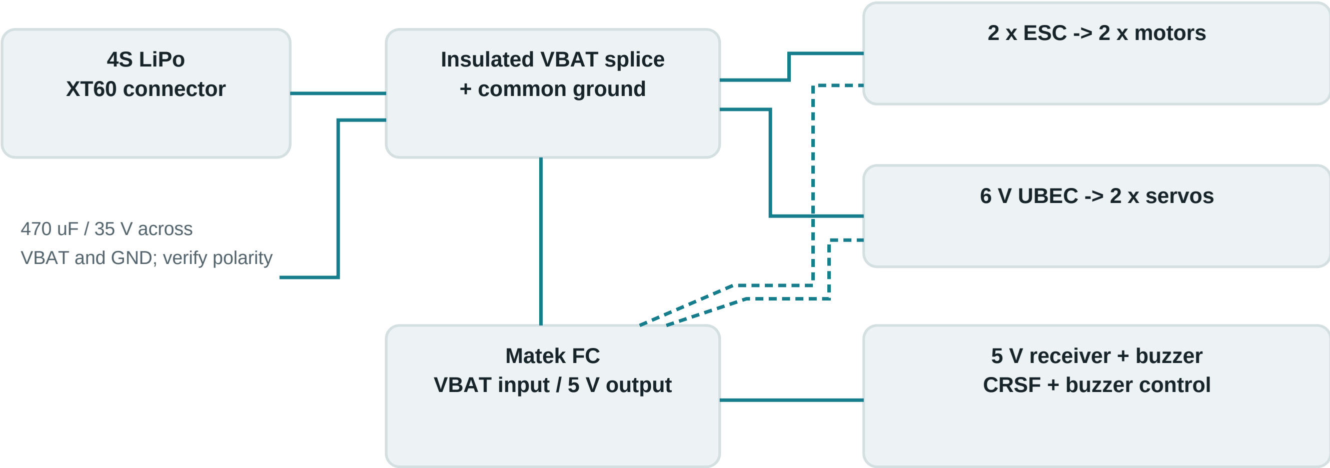
STL name (without .stl)	CAD g	Bounding box, mm
P11_L_bearing_yoke	10.64	18.0 x 74.0 x 35.0
P11_R_bearing_yoke	10.64	18.0 x 74.0 x 35.0
P12_L_motor_cradle	10.37	36.0 x 85.0 x 22.5
P12_R_motor_cradle	10.37	36.0 x 85.0 x 22.5
P14_L_servo_retainer	0.60	29.0 x 30.0 x 2.5
P14_R_servo_retainer	0.60	29.0 x 30.0 x 2.5
P15_L_ESC_tray	1.22	31.0 x 16.0 x 2.0
P15_R_ESC_tray	1.22	31.0 x 16.0 x 2.0
P16_Antenna_mast	0.10	5.0 x 5.0 x 10.0
P17_Shoulder_cowl_1	4.32	46.0 x 52.0 x 20.0
P17_Shoulder_cowl__1	4.32	46.0 x 52.0 x 20.0
P19_Tapered_tail_boom	22.77	150.0 x 56.6 x 64.5
P20_Servo_engine_cowl_1	4.33	63.0 x 30.4 x 21.0
P20_Servo_engine_cowl__1	4.33	63.0 x 30.4 x 21.0
P22_Cable_raceway_1	4.03	17.0 x 54.0 x 10.0
P22_Cable_raceway__1	4.03	17.0 x 54.0 x 10.0
P23_Main_wheel_1	0.64	14.0 x 4.0 x 14.0
P23_Main_wheel__1	0.64	14.0 x 4.0 x 14.0
P24_Tailwheel_fork	0.82	19.6 x 7.0 x 25.5
P25_Tailwheel	0.29	11.0 x 3.0 x 11.0

Optional guards - omitted from baseline

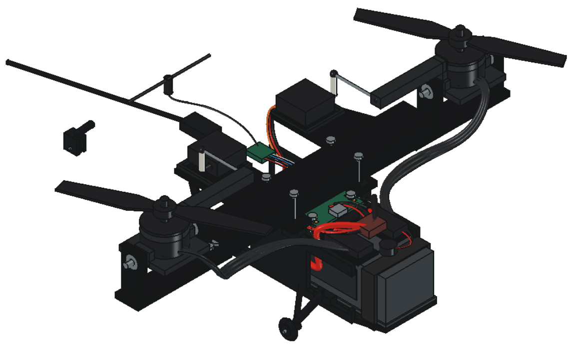
STL name	CAD g	Bounding box, mm
P13_L_tilting_guard	12.68	142.2 x 142.2 x 32.7
P13_R_tilting_guard	12.68	142.2 x 142.2 x 32.7

Installed printed mass: 250.90 g. Optional guards add 25.37 g. All 42 meshes fit the printer cube by bounding box; brim, supports and print orientation still need a slicer check.

Power & control wiring



Dashed: FC control signals. All grounds common; keep 5 V and 6 V positives separate.



Servo rail: dedicated regulated 6 V from UBEC. **Logic rail:** FC 5 V to receiver/buzzer. Grounds are common; do not join positive regulated rails. Main motor current bypasses the FC current sensor in this layout, so total pack current needs external measurement.

Proposed FC resources: S1/PC9 and S2/PC8 motors; S9/PB14 and S10/PA6 tilt servos. CRSF uses RX2/PA3 and TX2/PA2. Confirm actual firmware target, servo support, BI mixer, direction, neutral, endpoints and protocol with props removed. Older ESCs are not assumed to support DShot.

Wire schedule / 1 of 2

Neutral modeled paths. Cut estimate adds 15 mm for terminations and another 20 mm on each phase extension. Existing motor, servo and battery leads may replace extensions; hand dress and measure first. Endpoint names indicate function, not exact solder-pad geometry.

Wire	AWG	Route / cut mm	Endpoint functions
L_ESC_VBAT	18	22 / 37	Power splice + -> ESC L +
L_ESC_GND	18	17 / 33	Power splice - -> ESC L -
L_Phase_0	18	148 / 183	ESC L phase0 -> Motor L phase0
L_Phase_1	18	146 / 182	ESC L phase1 -> Motor L phase1
L_Phase_2	18	146 / 181	ESC L phase2 -> Motor L phase2
L_Servo_0	24	76 / 91	Servo L -> FC S9/S10
L_Servo_1	24	65 / 81	Servo L -> UBEC output
L_Servo_2	24	66 / 82	Servo L -> UBEC output
L_ESC_signal	26	30 / 45	FC S1/S2 -> ESC L signal
L_ESC_signal_ground	26	32 / 47	FC GND -> ESC L signal GND
R_ESC_VBAT	18	22 / 37	Power splice + -> ESC R +
R_ESC_GND	18	23 / 38	Power splice - -> ESC R -
R_Phase_0	18	148 / 183	ESC R phase0 -> Motor R phase0
R_Phase_1	18	146 / 182	ESC R phase1 -> Motor R phase1
R_Phase_2	18	146 / 181	ESC R phase2 -> Motor R phase2
R_Servo_0	24	76 / 91	Servo R -> FC S9/S10
R_Servo_1	24	65 / 81	Servo R -> UBEC output
R_Servo_2	24	66 / 82	Servo R -> UBEC output
R_ESC_signal	26	30 / 45	FC S1/S2 -> ESC R signal
R_ESC_signal_ground	26	32 / 47	FC GND -> ESC R signal GND
Battery_stock_positive	14	14 / 29	B1 pack + -> XT60 battery half +

42 routes are modeled and checked at neutral. Flexible travel, wire fatigue, wire-to-wire contact and every terminal/electronics pair are not continuously verified. Supplied antenna coax and pack balance leads are reference assemblies; do not fabricate or cut them from this table.

Wire schedule / 2 of 2

Neutral modeled paths. Cut estimate adds 15 mm for terminations and another 20 mm on each phase extension. Existing motor, servo and battery leads may replace extensions; hand dress and measure first. Endpoint names indicate function, not exact solder-pad geometry.

Wire	AWG	Route / cut mm	Endpoint functions
Battery_stock_negative	14	14 / 29	B1 pack - -> XT60 battery half -
B1_positive	14	121 / 137	XT60 + -> Power splice +
B1_negative	14	121 / 137	XT60 - -> Power splice -
UBEC_input_0	22	104 / 119	Power splice -> UBEC input
FC_input_0	26	35 / 51	Power splice -> FC battery input
Capacitor_0	24	114 / 129	Power splice -> C1
UBEC_input_1	22	113 / 128	Power splice -> UBEC input
FC_input_1	26	34 / 50	Power splice -> FC battery input
Capacitor_1	24	118 / 134	Power splice -> C1
Receiver_0	26	25 / 41	RP1 5V -> FC 5V
Receiver_1	26	25 / 41	RP1 GND -> FC GND
Receiver_2	26	25 / 41	RP1 TX -> FC RX2
Receiver_3	26	25 / 41	RP1 RX -> FC TX2
Balance_0	26	19 / 35	B1 balance tap -> J2 pin 1
Balance_1	26	19 / 35	B1 balance tap -> J2 pin 2
Balance_2	26	19 / 35	B1 balance tap -> J2 pin 3
Balance_3	26	19 / 35	B1 balance tap -> J2 pin 4
Balance_4	26	19 / 35	B1 balance tap -> J2 pin 5
Antenna_coax	30	75 / 90	RP1 UFL -> T antenna
Buzzer_wire_0	26	64 / 79	Buzzer -> FC buzzer pad
Buzzer_wire_1	26	64 / 79	Buzzer -> FC buzzer pad

42 routes are modeled and checked at neutral. Flexible travel, wire fatigue, wire-to-wire contact and every terminal/electronics pair are not continuously verified. Supplied antenna coax and pack balance leads are reference assemblies; do not fabricate or cut them from this table.

Lift, mass & control feasibility

Case	Mass g	TWR at 12 V	TWR at 16 V
Nominal	664.9	1.82	2.29
10% growth	731.4	1.65	2.09
Target ceiling	800	1.51	1.91

TWR = installed total static thrust / weight. The EMAX table covers original RS2205 2300KV motors with HQ5045BN two-blade props. An assumed 0.85 installed-thrust factor gives 1210 gf at 12 V and 1525 gf at 16 V, with a 24 A per-motor calculation ceiling where source data permit it. Neither factor nor limit is enforced firmware.

At nominal mass, simultaneous 20 degree pod tilt and 20 degree bank reduce vertical TWR to 1.61 at 12 V; at growth mass this falls to 1.46, before extra differential-command reserve. The growth case misses the preferred 1.8 vertical gate at low voltage. Fresh 16.8 V and hardware substitutes require new measurements.

Two motors are enough only with the tilt mechanism

Command	Bicopter response
Climb / descend	Both motors change total thrust.
Roll / lateral translation	Differential motor thrust rolls the aircraft; bank redirects thrust.
Pitch / forward-back motion	Coordinated common pod tilt and body attitude redirect thrust.
Yaw / turn heading	Opposite pod tilt creates a yaw moment.

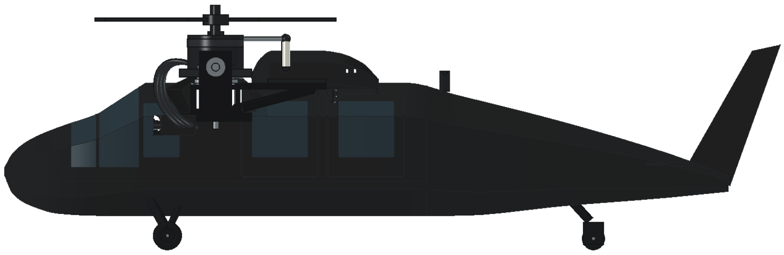
The local ideal model has sufficient control inputs, but pitch and yaw share servo travel: |common| + |differential| must stay within the absolute pod limit. TWR and controllability do not establish closed-loop stability.

Servo and endurance limits

BMS-115WV+ catalog torque is 5.5 kgf cm at 6 V; the audit uses one-quarter of that only as an assumed working screen. A bounded 45 deg/s roll gives approximately 0.0735 N m total servo load versus a 0.1348 N m screen. At 180 deg/s the screen is exceeded. Real rotating inertia, RPM, heat, friction, linkage flex and loaded response remain unknown.

Estimated hover endurance is roughly 3-4 minutes with 80% of 4S1300 nominal energy and a 5 W avionics/servo allowance. This is a static estimate, not measured endurance or a landing timer. See the full independent audit for equations and sensitivity cases.

Tail / balance and crosswind



Tail prints total **30.86 g**, already included in the 665 g estimate. The current calculated CG is **0.82 mm aft of the rotor pivot plane**. Removing the tail parts without rebalancing would shift CG to 7.28 mm forward. Hardware and finish remain in shared allowances.

The long tail accounts for about 43% of the coarse pitch inertia estimate and 25% of yaw inertia. It therefore changes response even though it is only about 5% of total mass. The passive tail does not need a tail rotor and does not inherently prevent bicopter flight.

Uniform side-flow sensitivity - no approved wind limit

Boom + fin summed side area is 71.38 cm^2, with a 179 mm area-weighted arm behind CG. Use N = 0.5 rho V^2 C integral[(CGx-x) dA], rho=1.225 kg/m^3. C=1.0-1.5 is an arbitrary sensitivity range, not measured or bounded aerodynamic data. Fin/boom overlap is double-counted; other aircraft surfaces and interference are excluded.

Side flow	Tail yaw moment, C=1.0-1.5	Ideal yaw candidate used
3 m/s	0.007-0.011 N m	6-9%
5 m/s	0.020-0.029 N m	17-26%
8 m/s	0.050-0.075 N m	44-66%
10 m/s	0.078-0.117 N m	69-103%

The comparison uses about 0.114 N m ideal yaw authority near hover, after reserving common trim within an unqualified +/-10 degree absolute pod-angle candidate. It assumes coordinated thrust and no simultaneous maneuver demand. At the higher sensitivity cases the tail alone can consume most or all of that candidate authority.

Conclusion: mass and balance are plausible with the tail, but crosswind, pitch loads, vibration and tail-joint strength still need measurement. No wind speed in this table is a safe or approved operating limit. Full method and reproducible calculations: engineering/TAIL_ASSESSMENT.md.

Printing & assembly sequence

Start with fit coupons, then structural pieces

Centauri Carbon basis: 256 mm cube. All STL bounds fit, but supports, brim and orientation still require review. Print bearing, screw, shell seam and servo-retainer fit coupons before committing the complete aircraft.

Use actual material density when weighing the build. The current shell uses a 1 mm section-coordinate inset, not a guaranteed constant surface-normal wall. Layer orientation, infill, heat, creep and fatigue need qualification; no universal slicer recipe is asserted to be sufficient.

1. Resolve the hardware

Measure actual battery, motors, ESCs, servos and FC. Finalize trunnion shoulders/spacers/retainers, spline fit, motor screw engagement and tail splice. The existing envelopes are provisional.

2. Assemble the structure

Dry-fit keel, crossbeam, avionics deck and yokes. Install metal bearings and positively retained pins. Check endplay and independent pod rotation.

3. Fit actuators and electronics

Set verified mechanical neutral, fit horns/links and check range. Install FC isolation, receiver, dedicated servo BEC, ESCs, buzzer and capacitor with service access.

4. Route and inspect wiring

Follow net functions and measure actual cut lengths. Insulate the splice and provide strain relief. Check phase-wire bend radius throughout travel; neutral CAD fit does not establish flex life.

5. Close the body and tail

Resolve the x=-105 mm tail splice and tailwheel/gear retention before fitting shell and glazing. Check cooling, antenna placement and disconnect access.

6. Weigh and rebalance

Measure completed mass and three-axis CG with battery. Aim for CGx near zero using battery position; rerun the audit if geometry, hardware or weight changes.

Detailed task list: docs/BUILD_AND_TEST.md. This sequence organizes prototype assembly; unresolved joints prevent it from being a fully released manufacturing instruction.

Verification & remaining gates

Completed computer checks	Scope / practical limit
167 valid physical CAD shapes	Topological validity; not stress, fatigue or actual hardware fit.
42 closed print meshes	40 installed, 2 optional; bounding box within printer limits.
22 sampled rotor poses	Selected fixed-part pairs; minimum prop clearance 9.906 mm. No continuous or deflected motion proof.
42 neutral wire routes	179,487 finite samples; no detected hits in completed scope. Not a flexible-harness or complete terminal-contact proof.
Mass, thrust and control screens	Source-based calculations and idealized models. No measured powertrain or tested flight tune.

Required before flight qualification

• Props removed: continuity, polarity, separated regulated rails, firmware mixer/resources, motor/servo direction, endpoints, neutral and loss-of-signal behavior.

• Loaded electrical tests: simultaneous servo current, BEC sag, capacitor behavior and temperatures. Check the actual ESC protocol. Main pack current needs external measurement in this layout.

• Restrained propeller-on tests: exact installed thrust/current over actual pack voltage, motor/ESC temperature and body interference. Fresh-pack operation needs evidence.

• Mechanical tests: layer/joint strength, crank flex, tail splice proof, pin/axle retention, vibration, cyclic durability and real harness movement.

• Dynamic tests: loaded servo tracking including gyroscopic loads; tail crosswind and pitch moments; actual controller response and reserve under simultaneous commands.

The +/-10 degree pod/bank and <=45 deg/s body-rate values are investigation candidates, not implemented settings or approved limits. No PID tune, full-flight simulation, CFD or physical flight test has been completed.

Status: flight_release = false. Keep measured configuration, firmware hash/settings, scale/CG results, thrust curves, thermal logs and pass/fail evidence together. A valid solid and TWR above one do not alone establish reliable flight.

References & project files

Primary specification references underpin the selected components. Exact battery SKU and generic hardware remain open. Reference pages are not a promise of current availability or endorsement. Full source snapshots and pinned Betaflight revisions are included in source/.

[MOTOR] EMAX RS2205 and test chart

[ESC] EMAX motor / Lightning ESC combination

[SERVO] Blue Bird BMS-115WV+ catalog

[FC] Matek F405-TE specifications and pinout

[BEC] Hobbywing UBEC 5A

[RX] RadioMaster RP1

[BATTERY] Tattu battery family reference - not a selected SKU

[PRINTER] Elegoo Centauri Carbon

[MIXER] Betaflight mixer documentation

[AERO] NASA drag equation

[COEFFICIENT] NASA coefficient and flow-condition limitations

NASA references support the force equation and coefficient limitations; the numerical tail areas, coefficients and moments are this project's own explicitly limited calculations.

Where to find the detail

Folder / file	Contents
cad/	Native FreeCAD model and STEP assembly
stl/	Installed printable pieces; optional guards in subfolder
engineering/	Full audits, tail assessment, wire schedule, calculations and geometry-check evidence
docs/	Purchase/print list, machine-readable parts and build/test plan
scripts/	Geometry, analysis, audits, documentation and validation generators
REPRODUCE.md	Commands, dependencies and limits of reproducibility
NOTICE.md	Manufacturer references and third-party source attribution

Repository: github.com/ar3116-sketch/kestrel-bicopter